Homework Solutions

Math 1A

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Problem Q1

Can the graph of a function have more than one tangent at a given point? If so, graph your answer. If not, explain why.

Solution. In general, a smooth function has exactly one tangent line at each point.

However, a curve (not necessarily the graph of a function) can have multiple tangents at a self-intersection point.

For a function graph, the tangent is unique where the derivative exists.

Answer:

$$\boxed{\text{Unique tangent where } f'(a) \text{ exists.}}$$

Problem Q2

Is there a function whose graph doesn't have a tangent at some point? If so, graph your answer. If not, explain why.

Solution. Yes. A function may fail to have a tangent at a point where it is not differentiable.

Examples: $f(x) = |x|$ at $x = 0$ (corner), $f(x) = x^{1/3}$ at $x = 0$ (vertical tangent/cusp).

Answer:

$$\boxed{\text{Yes: } f(x) = |x| \text{ at } x = 0 \text{ (corner)}}$$

Problem P1

Suppose δ is a positive real number. How do you describe all real numbers x that are within δ of 0?

Solution. x within of 0: $|x| < \delta$, i.e. $-\delta < x < \delta$

Answer: $|x| < \delta$, i.e. $-\delta < x < \delta$

(a) The set of all x within δ of 0: $-\delta < x < \delta$ $\boxed{-\delta < x < \delta}$

(b) $|x| < \delta$ $\boxed{|x| < \delta}$

(c) $(-\delta, \delta)$ $\boxed{(-\delta, \delta)}$

Problem P2

Suppose a is a real number and δ is a positive real number. How do you describe all real numbers x that are within δ of a?

Solution. x within of a: $|x-a|<\delta$, i.e. $a-\delta<x<a+\delta$

Answer: $|x-a|<\delta$, i.e. $a-\delta<x<a+\delta$

(a) [Number line with open circles at endpoints, center marked] See graph

(b) $a-\delta<x<a+\delta$ Equivalently: $-\delta<x-a<\delta$ $\boxed{a-\delta<x<a+\delta}$

(c) $|x-a|<\delta$ $\boxed{|x-a|<\delta}$

(d) $(a-\delta,a+\delta)$ $\boxed{(a-\delta,a+\delta)}$

Problem P3

Suppose a is a real number and δ is a positive real number. How do you describe all real numbers x that are within δ of a, but not equal to a:

Solution. x within of a, not equal to a: $0<|x-a|<\delta$

Answer: $0<|x-a|<\delta$

(a) [Number line with open circles at endpoints, center marked] See graph

(b) $a-\delta<x<a+\delta, \quad x\neq a$ Or equivalently: $a-\delta<x<a$ or $a<x<a+\delta$ $\boxed{a-\delta<x<a+\delta, \quad x\neq a}$

(c) $0<|x-a|<\delta$ $\boxed{0<|x-a|<\delta}$

(d) $(a-\delta,a)\cup(a,a+\delta)$ $\boxed{(a-\delta,a)\cup(a,a+\delta)}$

Problem P4

Let $f(x)=x^2$.

Solution. Given: $f(x)=x^2$

Answer:

$$\boxed{f(x)=x^2}$$

(a) Need: $9.0-1<x^2<9.0+1$ i.e. $8.0<x^2<10.0$ Solving: $x\in(2.82842712474619, 3.162277660168$ (positive x only) $\boxed{(2.82842712474619, 3.16227766016838)}$

(b) Need: $|x-3|<\delta \Rightarrow |x^2-9.0|<1$ Boundaries: $x^2=10.0 \Rightarrow x=[-3.16227766016838,\ 3.1622776$ $x^2=8.0 \Rightarrow x=[-2.82842712474619,\ 2.82842712474619]$ Minimum distance from $x=3$: $\delta=0.16227766016838$ $\boxed{\delta=0.16227766016838}$

(c) -

(d) Yes, this is true. For any $\varepsilon>0$, we can find $\delta>0$ such that $|x-a|<\delta \Rightarrow |x^2-L|<\varepsilon$. This follows because $f(x)$ is continuous at $x=a$. $\boxed{\text{Yes (continuity)}}$

Problem AP1

Graph $y = 2x - x^2$. For which points a is the tangent line to the curve at the point $(a, 2a - a^2)$ a horizontal line? How is the value of $2x - x^2$ at these points related to the values of $2x - x^2$ at other points?

Solution. Given: $y = -x^2 + 2x$
Derivative: $y' = 2 - 2x$
Horizontal tangent where $y' = 0$: $2 - 2x = 0$
Solutions: $x = [1]$
Points: $(1, 1)$
$f''(1) = -2 < 0$: this is a maximum.
The value 1 is the maximum of $-x^2 + 2x$.
Answer:

$$\boxed{(1, 1)}$$

Problem AP2

The point $(2, 0)$ is on the curve $y = 2x - x^2$.

Solution. Given: $y = -x^2 + 2x$
Derivative: $y' = 2 - 2x$
At point $(2, 0)$:
Slope: $f'(2) = -2$
Tangent line: $y = 4 - 2x$
Answer:

$$\boxed{4 - 2x}$$

(a) Draw line through $(2, 0)$ with slope -2 $\boxed{\text{Line with slope } -2}$

(b) Slope at $x = 2$: -2 Tangent: $y - 0 = -2(x - 2)$ $y = 4 - 2x$ $\boxed{y = 4 - 2x}$

Auto-generated: 8/8 problems solved by SymPy.